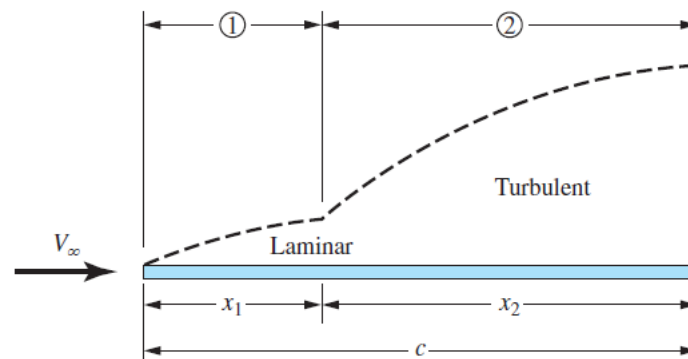


Aerodynamics Tutorial

Given an airfoil with a chord length of 1.5 m and a Reynolds number 3.1×10^6 provided, utilize the laminar and turbulent boundary layer equations on a smooth flat plate to estimate the skin-friction drag coefficient of the NACA 2412 airfoil at a zero-degree angle of attack. The measured drag coefficient of the airfoil is 0.0068. Analyze and provide your thoughts on the drag coefficient data and determine the pressure drag coefficient.

Solution:



The critical Reynolds number is given as,

$$Re_{cr} = \frac{\rho_\infty U_\infty x_{cr}}{\mu_\infty} = 5 \times 10^5$$

The Reynolds number based on chord length is given as

$$Re_{cr} = \frac{\rho_\infty U_\infty c}{\mu_\infty} = 3.1 \times 10^6$$

From the critical Reynolds number and Reynolds number based on chord length, we get

$$\frac{x_{cr}}{c} = \frac{x_1}{c} = \frac{5 \times 10^5}{3.1 \times 10^6} = 0.1613$$

Important to Note: Because the Reynolds number in the equations for skin friction drag coefficient is always based on length measured *from the leading edge*, we cannot simply calculate the turbulent skin friction drag coefficient for region 2.

Assuming all turbulent flow over the entire length of the plate, the drag (on one side of the plate) is

$$D_{c,turbulent} = q_\infty S (C_{f,c})_{turbulent} = q_\infty (c)(1) (C_{f,c})_{turbulent}$$

$S = (c)(1)$ per unit span

The turbulent drag on just Region 1 is

$$D_{1,\text{turbulent}} = q_{\infty} S (C_{f,1})_{\text{turbulent}} = q_{\infty} (x_1)(1)(C_{f,1})_{\text{turbulent}}$$

Thus, the turbulent drag just on Region 2 is

$$D_{2,\text{turbulent}} = D_{c,\text{turbulent}} - D_{1,\text{turbulent}}$$

$$D_{2,\text{turbulent}} = q_{\infty} c (C_{f,c})_{\text{turbulent}} - q_{\infty} x_1 (C_{f,1})_{\text{turbulent}}$$

The laminar drag on Region 1 is

$$D_{1,\text{laminar}} = q_{\infty} S (C_{f,1})_{\text{laminar}} = q_{\infty} (x_1)(1)(C_{f,1})_{\text{laminar}}$$

The total skin-friction drag on the flat plate is then

$$D_{\text{total}} = D_{1,\text{laminar}} + D_{2,\text{turbulent}} = q_{\infty} x_1 (C_{f,1})_{\text{laminar}} + (q_{\infty} c (C_{f,c})_{\text{turbulent}} - q_{\infty} x_1 (C_{f,1})_{\text{turbulent}})$$

Skin-friction coefficient for laminar BL is

$$C_{f,\text{laminar}} = \frac{1.328}{\sqrt{\text{Re}}}$$

Similarly, the skin friction coefficient for Turbulent BL is

$$C_{f,\text{turbulent}} = \frac{0.074}{\text{Re}^{(1/5)}}$$

The Reynolds number for Region 1 is

$$\text{Re}_{x_1} = \frac{\rho_{\infty} U_{\infty} x_1}{\mu_{\infty}} = \frac{\rho_{\infty} U_{\infty} x_1}{\mu_{\infty}} \left(\frac{c}{c} \right) = \left(\frac{\rho_{\infty} U_{\infty} c}{\mu_{\infty}} \right) \left(\frac{x_1}{c} \right) = \text{Re}_c \left(\frac{x_1}{c} \right) = 3.1 \times 10^6 \times 0.1613 = 5 \times 10^5$$

The skin-friction coefficient for Region 1 (**laminar**) is

$$C_{f,\text{laminar}} = \frac{1.328}{\sqrt{\text{Re}}} = \frac{1.328}{\sqrt{5 \times 10^5}} = 0.00188$$

The skin-friction coefficient for Region 1 (**turbulent**) is

$$C_{f,turbulent} = \frac{0.074}{Re^{(1/5)}} = \frac{0.074}{(3.1 \times 10^6)^{0.2}} = 0.00372$$

The skin-friction coefficient for the entire plate (**turbulent**) is

$$C_{f,turbulent} = \frac{0.074}{Re^{(1/5)}} = \frac{0.074}{(5 \times 10^5)^{0.2}} = 0.00536$$

The skin-friction coefficient for the entire plate is

$$C_{f,total} = \frac{D_{total}}{q_{\infty} S} = \frac{D_{total}}{q_{\infty} (c)(1)} = \frac{q_{\infty} x_1 (C_{f,1})_{laminar} + q_{\infty} c (C_{f,c})_{turbulent} - q_{\infty} x_1 (C_{f,1})_{turbulent}}{q_{\infty} c}$$

$$C_{f,total} = \left(\frac{x_1}{c} \right) (C_{f,1})_{laminar} + (C_{f,c})_{turbulent} - \left(\frac{x_1}{c} \right) (C_{f,1})_{turbulent}$$

$$C_{f,total} = (0.1613)(0.00188) + 0.00372 - (0.1613)(0.00536) = 0.003158$$

Considering both sides of the flat plate,

$$C_{f,total} = 2 \times 0.003158 = 0.0063$$

The measured airfoil drag coefficient is 0.0068, which includes *both* skin friction drag and pressure drag due to flow separation.

$$C_{d,pressure} = 0.0068 - 0.0063 = 0.0005$$

The pressure drag due to flow separation is about **7.4 percent** of the total drag.